RL Agent vs Baseline Methods
RL Agent
Best Possible Two-Qubit Gate Reduction
Best Possible Depth Reduction
Best Fixed Reward Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Optimizer (WISQ BQSKit)
Best Depth Reduction Chain (TKET --> Qiskit standard)
Best Depth Reduction (TKET)
0.355
0.342
0.274
0.227
0.227
0.124
0.080
0.079
0.00 0.05 0.10 0.15 0.20 0.25 0.30 0.35 0.40
Average utility reward
Best Possible Two-Qubit Gate Reduction
RL Agent
Best Fixed Reward Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Optimizer (WISQ BQSKit)
Best Depth Reduction Chain (TKET --> Qiskit standard)
Best Depth Reduction (TKET)
22.4%
20.1%
18.9%
18.9%
14.2%
6.3%
6.3%
0 5 10 15 20 25
Average two-qubit gate reduction (%)
Best Possible Depth Reduction
RL Agent
Best Depth Reduction Chain (TKET --> Qiskit standard)
Best Depth Reduction (TKET)
Best Fixed Reward Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Optimizer (WISQ BQSKit)
13.2%
8.6%
-4.3%
-5.7%
-16.6%
-16.6%
-36.3%
−30 −20 −10 0 10 20 30 40
Average depth reduction (%)
Best Fixed Reward Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Chain (TKET --> WISQ BQSKit)
Best 2Q Gate Reduction Optimizer (WISQ BQSKit)
RL Agent
Best Depth Reduction Chain (TKET --> Qiskit standard)
Best Depth Reduction (TKET)
145.71
145.71
119.25
81.57
25.32
23.79
0 20 40 60 80 100 120 140 160
Average runtime (s)